## Step 1: Find the last digit

The last digit of a number can be found by using the modulo operator (). The expression $n\%10$ wil give the remainder when $n$ is divided by 10, which is the last digit.

## Step 2: Find the first digit

To find the first digit, repeatedly divide the number by 10 until it becomes a single-digit number.

## Step 3: Write the C code

C

```c
#include <stdio.h>
int main() {
    int n, firstDigit, lastDigit;
    printf("Enter a number: ");
    scanf("%d", &n);
    lastDigit = n % 10;
    firstDigit = n;
    while (firstDigit >= 10) {
        firstDigit = firstDigit / 10;
    }
    printf("First digit: %d\n", firstDi
```

## Step 1: Understand the factorial formula

The factorial of a non-negative integer $n$, denoted by $n!$, is the product of all positive integers less than or equal to $n$. The formula

$$n! = n \times (n-1) \times (n-2) \times \ldots \times 1$$

For example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$ factorial of 0 is 1.

## Step 2: Write the C code

C

```
#include <stdio.h>
int main() {
    int n, i;
    unsigned long long factorial = 1
    printf("Enter a non-negative int
    scanf("%d", &n);
    if (n < 0) {
        printf("Error! Factorial of
```

## Step 1: Understand the definition c a prime number

A prime number is a natural number greater tl 1 that has no positive divisors other than 1 anc itself.

## Step 2: Write the C code

C

```c
#include <stdio.h>
int main() {
    int n, i, flag = 0;
    printf("Enter a positive integer:
    scanf("%d", &n);
    if (n == 0 || n == 1)
        flag = 1;
    for (i = 2; i <= n / 2; ++i) {
        if (n % i == 0) {
            flag = 1;
            break;
        }
    }
    if (flag == 0)
```

```c
#include <stdio.h>

long long factorial(int n) {
    long long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

int main() {
    int num, originalNum, digit;
    long long sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);

    originalNum = num;

    while (num > 0) {
        digit = num % 10;
        sum += factorial(digit);
        num /= 10;
    }

    if (sum == originalNum) {
```

## 5. Find sum of all digits in a number

### Step 1: Understand the process

To find the sum of all digits in a number, you can repeatedly extract the last digit using the modulo operator () and add it to a sum, then remove the last digit by integer division (/) until the number becomes 0.

### Step 2: Write the C code

C

```c
#include <stdio.h>
int main() {
    int n, temp, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
        sum += temp % 10;
        temp /= 10;
    }
    printf("Sum of digits of %d = %d\n",
    return 0;
```